

ORAL PRESENTATION

# Indirect Thermal-Aware Supervision for Agonist– Antagonist SMA Ankle Actuation

---

Sergio Jacobo-Zavaleta · Dorin Copaci · Sofía Die-Pancorbo · Dolores Blanco · Luis Moreno

Department of Systems Engineering and Automation · Universidad Carlos III de Madrid

# Motivation and control challenge

## Compact actuation is attractive, but heat recovery limits endurance

### Why SMA?

**Lightweight**, **silent** and compact actuation for early-stage ankle mobilization.

### Main limitation

Heating is fast, but **cooling is slow**; repeated activation can produce **heat accumulation**.

### Control challenge

Preserve **closed-loop tracking** while reducing **unnecessary Joule heating**.

The goal is not only accurate motion, but **sustained tracking** through control decisions that reduce unnecessary heating.

### Core idea

Do not only ask:

**“How much should I track?”**

Also ask:

**When should energy enter the actuator?**

Agonist-antagonist SMA

Free cooling

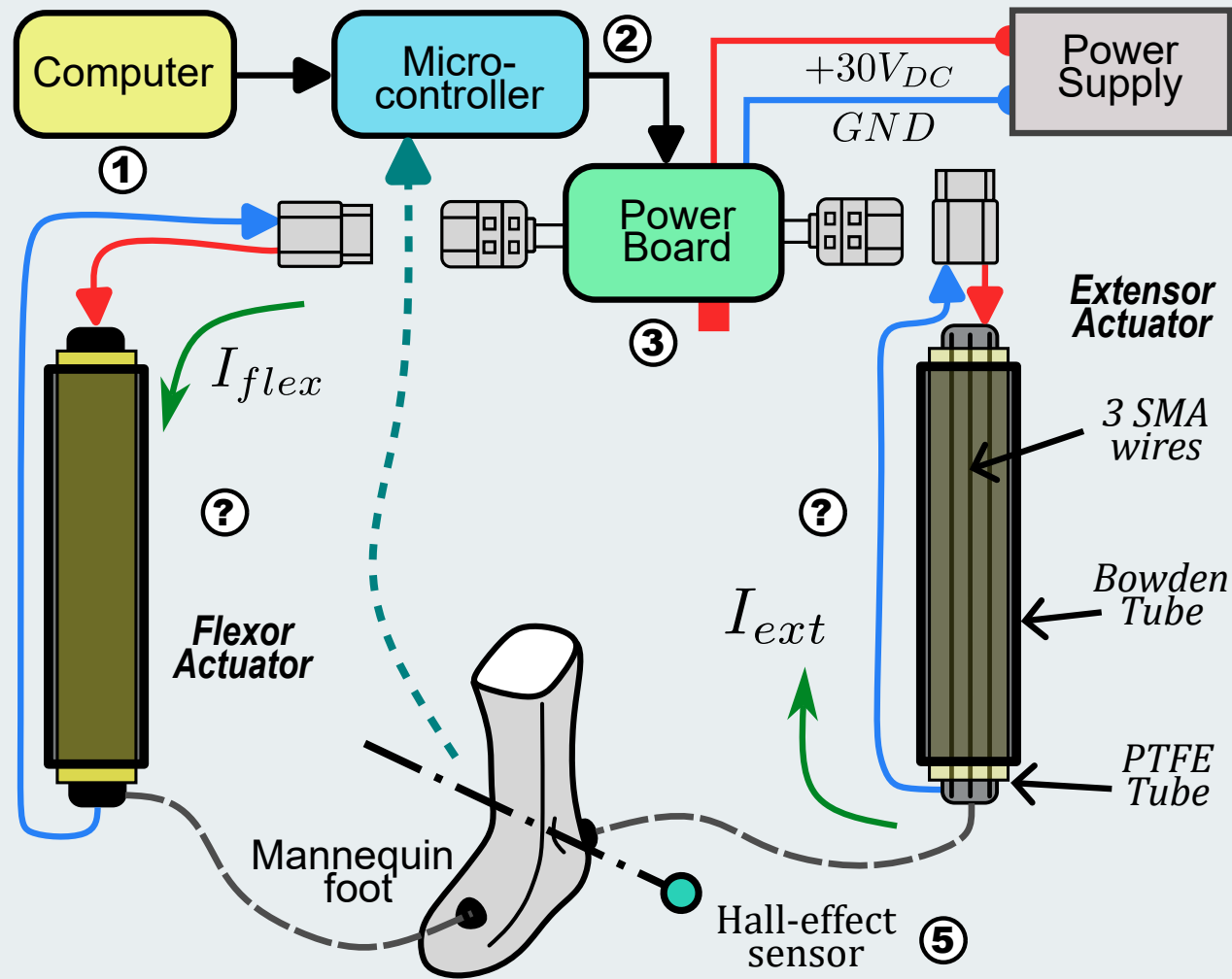
Thermal-aware supervision

BPID tracking

# SYSTEM

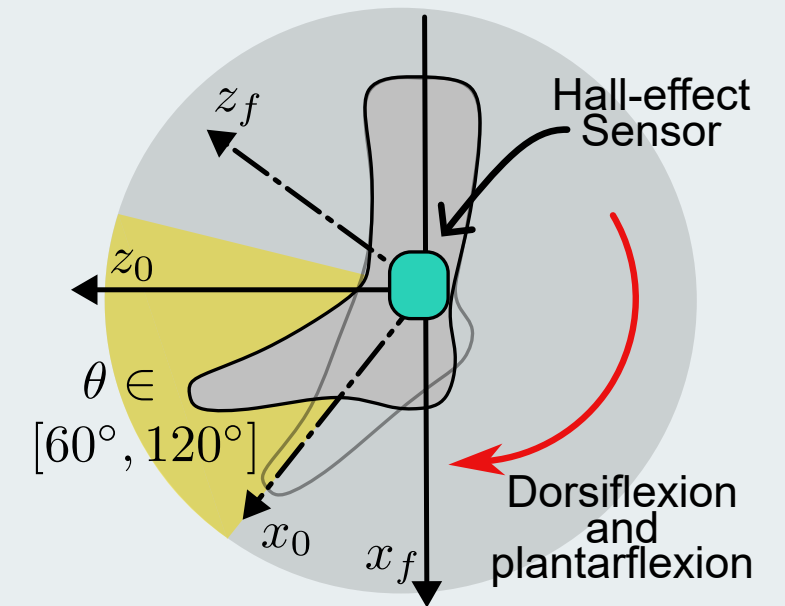
Antagonistic SMA ankle module

# Actuation architecture



Hardware-in-the-loop Setup

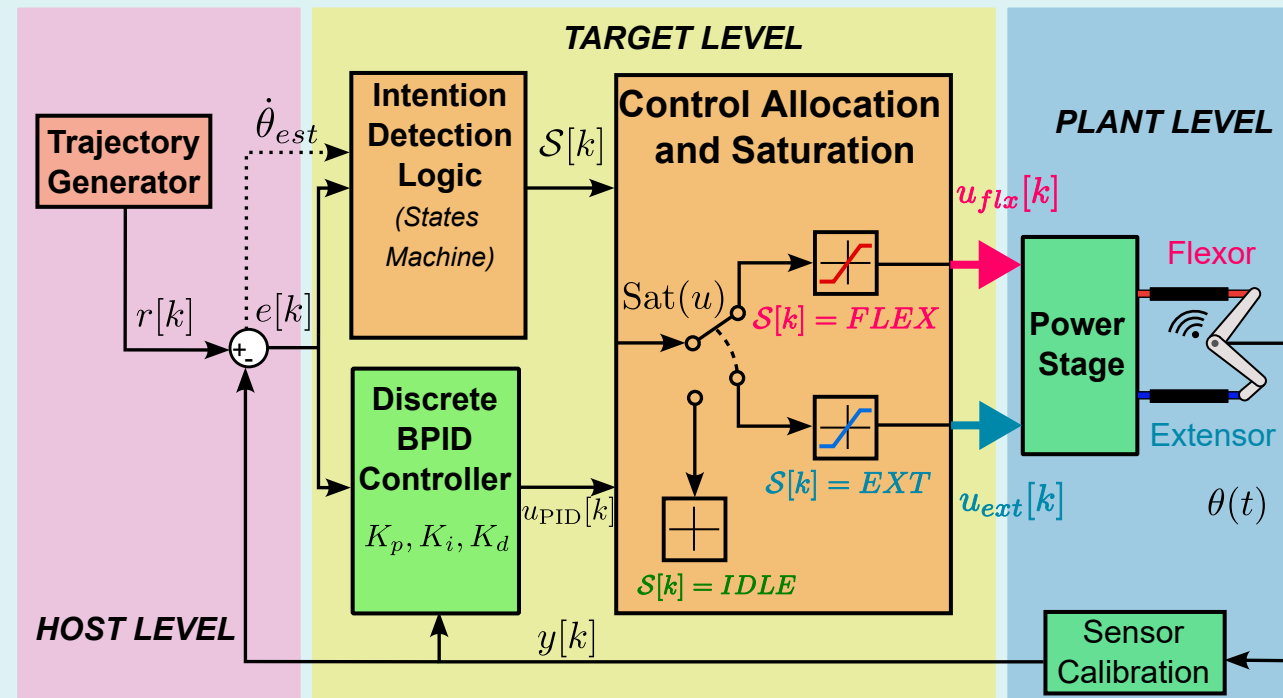
1. Host computer
2. STM32F4 target
3. Power electronics
4. Physical plant
5. Position sensor



# CONTROL

Tracking + thermal-aware supervision

# Closed-loop control scheme



*A two-layer hierarchical logic executed across three layers.*

The control scheme consists of:

## Supervisory decision layer

Determines when each SMA group is allowed to receive Joule heating.

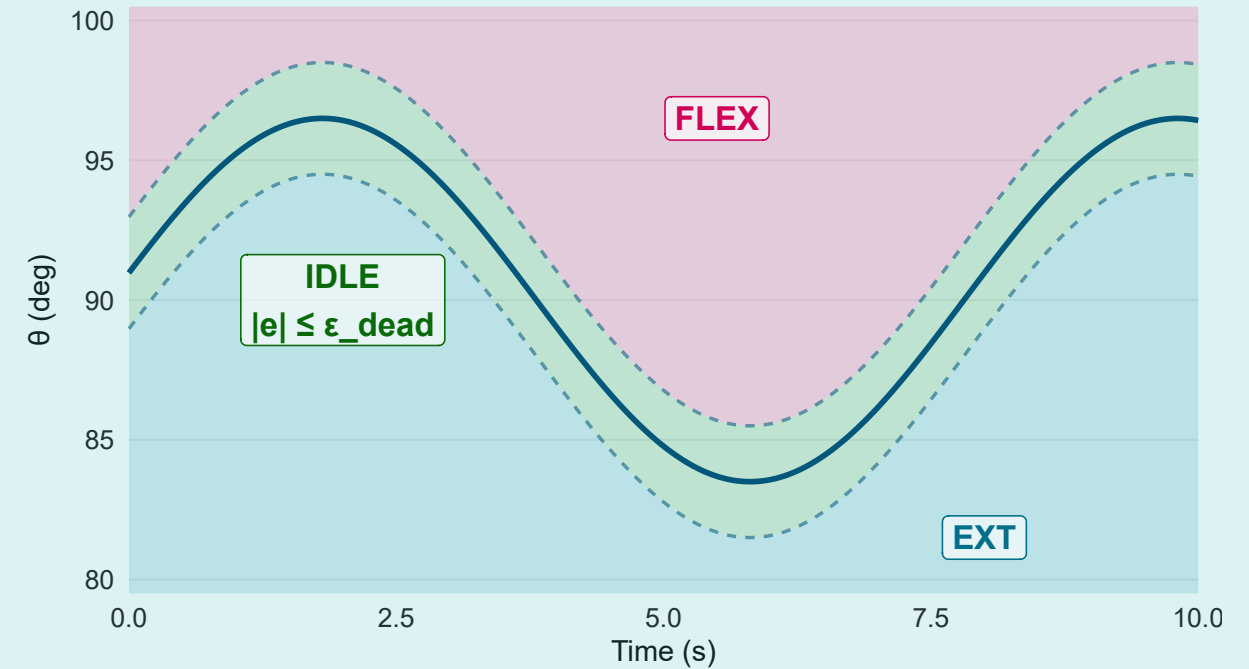
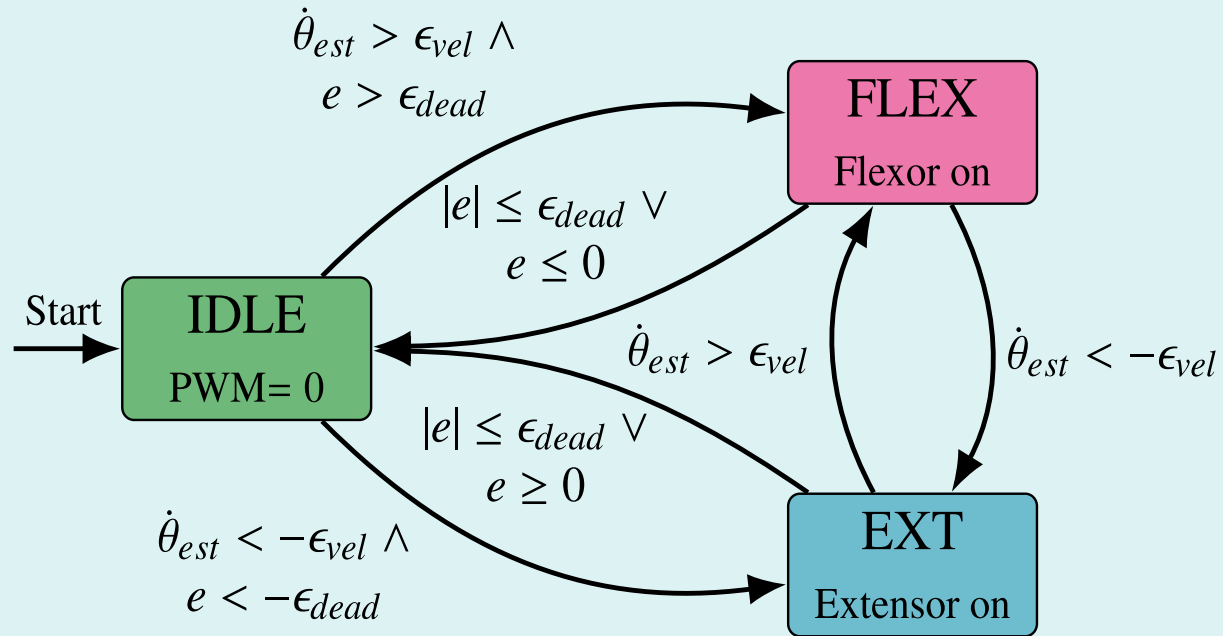
## Bilinear PID layer

Computes how much control effort is applied to the active group.

The **supervisor** decides **when to heat**; the **BPID** decides **how much**.

# State logic

## State machine (motion intention and tracking error)



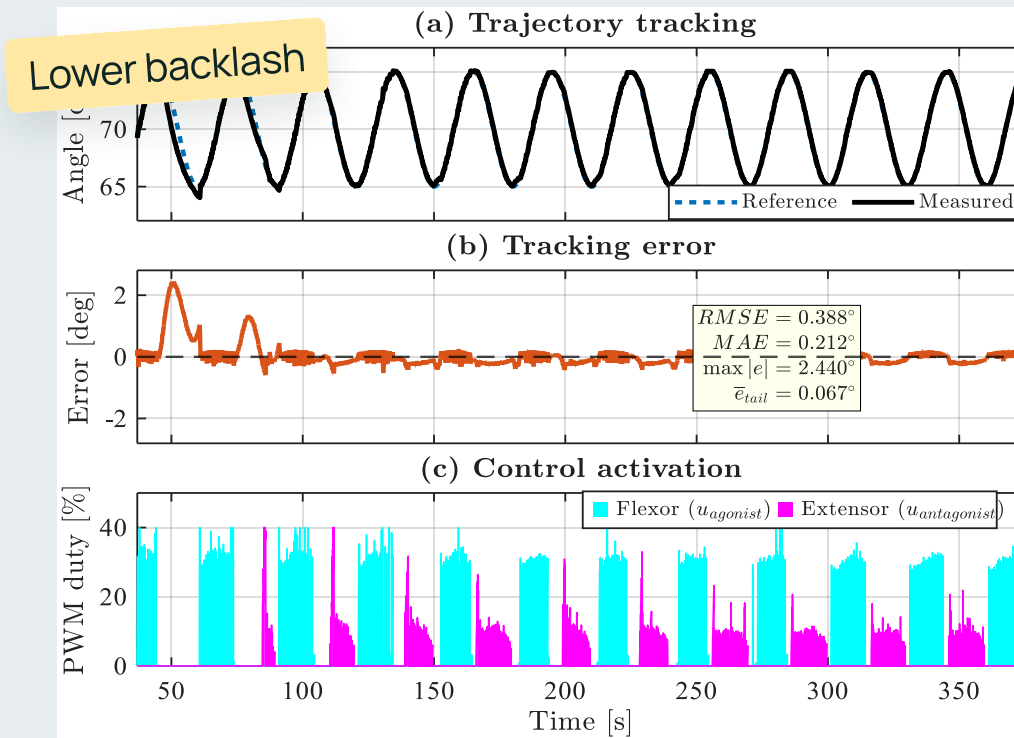
Conceptual plot of supervisory bands

Inside the band, the system stays in **IDLE** and exploits **free cooling**.

# RESULTS

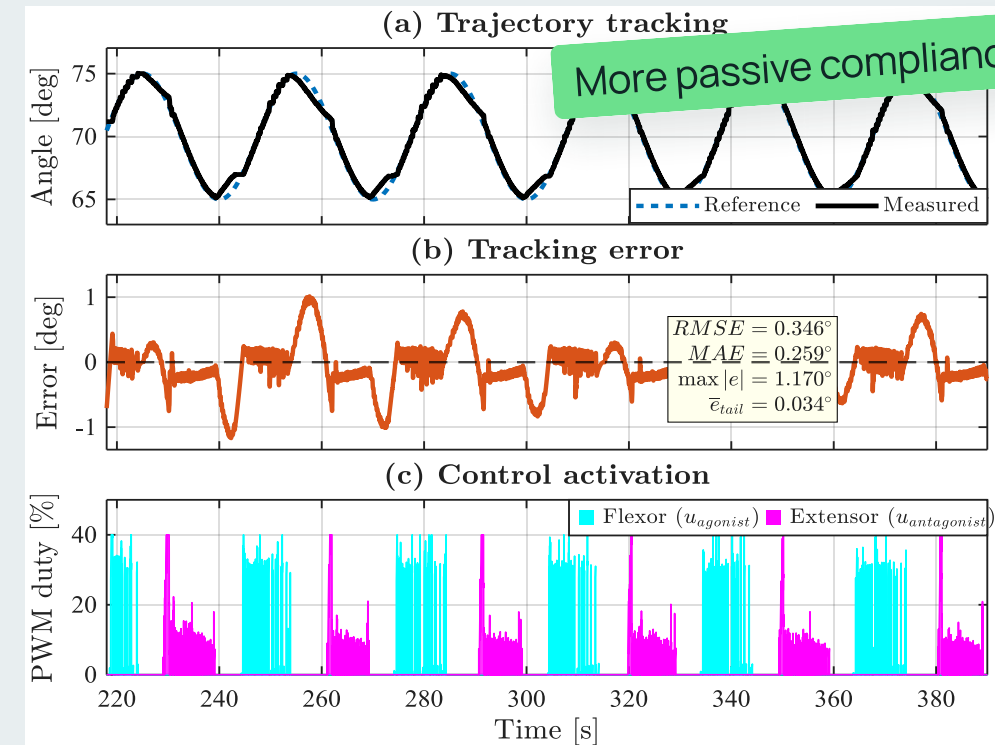


# Results: configuration comparison



*High pre-tension configuration*

$$\text{RMSE} = 0.388^\circ \cdot \text{MAE} = 0.212^\circ \cdot \max |e| = 2.440^\circ$$



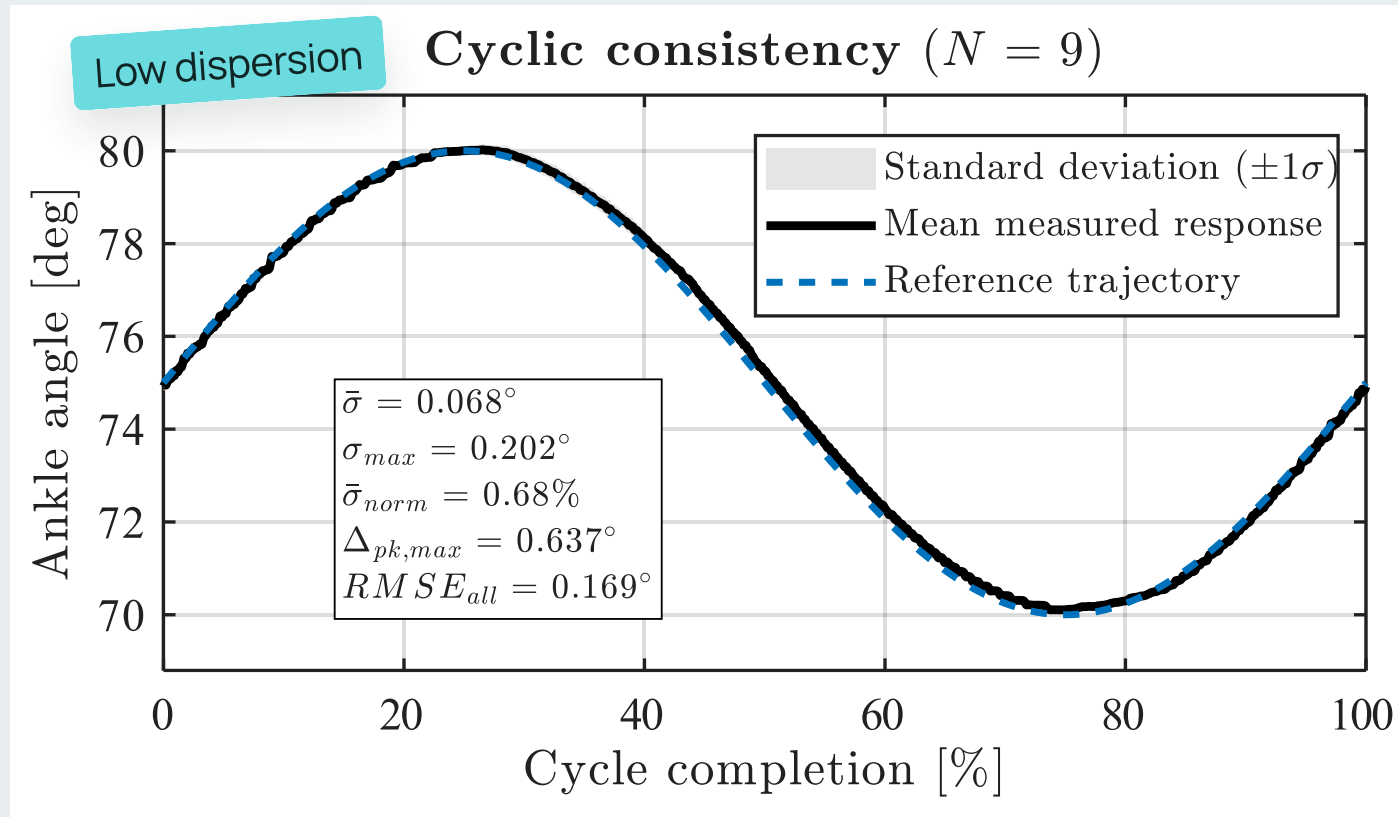
*Relaxed initial state configuration*

$$\text{RMSE} = 0.346^\circ \cdot \text{MAE} = 0.259^\circ \cdot \max |e| = 1.170^\circ$$

**High pre-tension** improves early mechanical response, but the **relaxed state** reduces peak error and promotes recurrent **free-cooling intervals**.

# Results: cyclic repeatability

The response remains consistent from cycle to cycle



Mean dispersion:  $\bar{\sigma} = 0.068^\circ$

Maximum dispersion:  $\sigma_{max} = 0.202^\circ$

Peak mismatch:  $\Delta_{pk,max} = 0.637^\circ$

Overall cyclic RMSE:  $RMSE_{all} = 0.169^\circ$

This indicates:

High repeatability

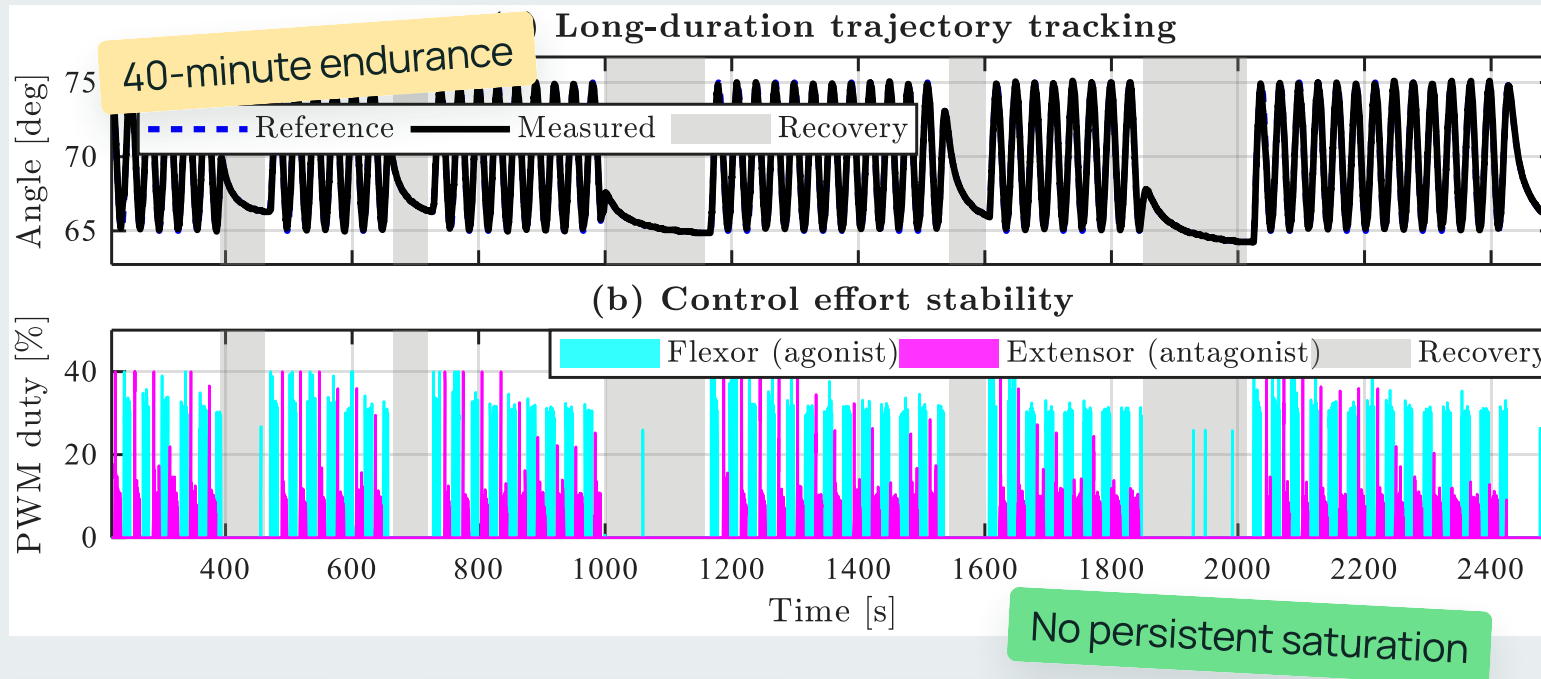
Low inter-cycle drift

Stable closed-loop behavior

The **small dispersion band** shows that tracking is not only accurate, but **repeatable cycle after cycle**.

# Results: long-duration endurance

## Stable operation over 40 minutes without thermal saturation



### Duration

40 min closed-loop operation

### Tracking accuracy

$\text{RMSE}_{\text{active}} = 0.218^\circ$

### Worst-case error

$\max |e| = 1.384^\circ$

### Thermal behavior

Recovery intervals prevent persistent heat accumulation

Recovery intervals

No tracking collapse

Closed-loop recovery

The key result is **sustained tracking**: recovery windows are embedded in the control strategy, preventing persistent heat accumulation while keeping the loop active.

# Discussion and take-home message

## What do these results mean?

**Tracking accuracy alone is not enough**: in SMA systems, **thermal recovery** must be part of the control architecture.

The supervisor does not need direct temperature sensing. It uses **motion intention** and **signed tracking error**.

**Inside the band**, the controller stays in **IDLE** and exploits **free cooling**.

### Take-home message

**Indirect thermal-aware supervision** enables SMA ankle actuation that is **and thermally robust**.

# Thank you

DISCUSSION

## Questions?

---

Indirect Thermal-Aware Supervision for Agonist–Antagonist SMA Ankle Actuation

Sergio Jacobo-Zavaleta · Dorin Copaci · Sofía Die-Pancorbo · Dolores Blanco · Luis Moreno